Data sources

Data ingestion

- **Batch data** — Scheduled · local · cloud
- **Real-time streaming** — Kafka · Kinesis · Pub/Sub
- **Change data capture** — Debezium · CDC
- **Reverse ETL** — Warehouse → ops systems

Data Ocean

- **Data lake** — Hot · warm · archive
- **Data pond** — Domain store
- **Data puddle** — Project-level
- **Data mesh** — Domain ownership
- **Data contracts** — Schema · SLA · owner
- **Workflow manager**

Data transformation

- **Data labeling** — Human + auto labeling
- **Query engines** — Spark · Trino · BigQuery
- **Data sci. libraries** — Pandas · NumPy · dbt
- **Data enrichment** — External signal ingestion

AI infrastructure

Knowledge store

- Feature store
- Vector database
- Knowledge graph

LLMOps + MLOps

- Model registry
- Prompt registry
- Fine-tuning · RLHF
- Eval & red-teaming
- CI/CD · experiment tracking

AI inference

- Online predictor
- Batch predictor
- Streaming inference
- Edge inference
- AI gateway · model routing

GenAI platform

- **Guardrails & orchestration** — Safety filters · LangGraph
- **Model & prompt hub** — Versioned prompts · evals
- **Custom knowledge model** — Domain fine-tuning
- **Foundation models** — Large · small · open-source
- **RAG pipeline** — Retrieval-augmented gen.
- **Multimodal** — Vision · audio · document

Agentic AI layer

- **Agent orchestrator** — CrewAI · AutoGen · LangGraph
- **Tool registry / MCP** — APIs · code · search tools
- **Agent memory** — Episodic · semantic · working
- **HITL interface** — Human approval gates
- **Agent monitoring** — Loops · cost · policy guards

Analytics & BI

- **Data viz & storyboard** — Dashboards · reports
- **Data search** — Semantic · full-text
- **Data accumulation** — Aggregation · caching
- **Content analysis** — NLP · classification
- **Decision management** — Rules · optimisation
- **Agentic analytics** — AI-driven insight gen.

AI trust, governance & security

Data governance
Data lineage · Master & entity data · Data anonymization · Data authentication · Data quality · Data catalog · Data protection · Security intelligence

AI governance
AI policies · AI model control · AI leakage testing · AI trusted data · Agent audit trails · Responsible AI dashboard · AI usage control · LLM guardrails

Assessment framework

- **AI maturity assessment**
- **Economic benefit assessment**
- **Responsible AI KPIs**

Applications

Data layers · AI/ML infrastructure · GenAI & analytics · Agentic AI · Governance & trust · Assessment